

Bounded Convolution Kernels on Locally Compact Abelian Groups

A complete LCA subcase of questions of Forrest–Spronk–Wood

Autonomous proof candidate for human review

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Status. Candidate partial result, likely valid. The theorem completely settles two of the source questions for locally compact abelian groups. It does not settle the questions for arbitrary locally compact groups and does not fully describe the closure in the negative cases.

1 Source question

Let G be a locally compact group. Forrest, Spronk, and Wood [1] write $\text{Conv}^\infty(G)$ for the bounded measurable functions φ whose left-convolution operator extends boundedly to $L^2(G)$, and denote that operator by $\Lambda(\varphi)$. On page 8, at the end of Section 2.1, they ask for which groups

$$\text{Conv}^\infty(G) \subseteq L^2(G),$$

when $\overline{\Lambda(\text{Conv}^\infty(G))}^{\|\cdot\|}$ is a $*$ -subalgebra of $\text{VN}(G)$, and how large that closure is.

Unfortunately, we cannot determine if the formula (2.3) obtains for any u in $A(G)$, since if $u \notin S^1A(G)$, then the limit of the integrals in cannot be expected to hold. $\square$

We note that if G is discrete, then it is well-known that every element of $\text{VN}(G)$ is an ℓ^2 -convolver, and in particular an ℓ^∞ -convolver, i.e. $\text{VN}(G) = \Lambda(\text{Conv}^\infty(G))$. This corresponds to the fact that $S^1A(G) = A(G) \cap \ell^1(G) = \ell^1(G)$.

If G is compact, then $L^\infty(G) \subseteq L^1(G)$, and hence $L^\infty(G) = \text{Conv}^\infty(G)$. This corresponds to the fact that $S^1A(G) = A(G)$, in this case. Here we see that $\Lambda(\text{Conv}^\infty(G))$ is a dense $*$ -subalgebra of the (reduced) group C^* -algebra $C_r^*(G)$.

There are natural questions concerning the extent of $\text{Conv}^\infty(G)$. We note that in both of the above cases that $\text{Conv}^\infty(G) \subseteq L^2(G)$. Moreover, we have that $L^\infty(G) \cap L^1(G)$, which is necessarily always contained in $\text{Conv}^\infty(G)$, is always a subset of $L^2(G)$. Thus we ask: *for which G is $\text{Conv}^\infty(G) \subseteq L^2(G)$?* It is also natural to wonder about the norm closure $\overline{\Lambda(\text{Conv}^\infty(G))}^{\|\cdot\|}$. *When is $\overline{\Lambda(\text{Conv}^\infty(G))}^{\|\cdot\|}$ a $*$ -subalgebra of $\text{VN}(G)$?* *How big is $\overline{\Lambda(\text{Conv}^\infty(G))}^{\|\cdot\|}$?* We note that $\overline{\Lambda(\text{Conv}^\infty(G))}^{\|\cdot\|}$ always contains the reduced C^* -algebra $C_r^*(G)$ of G .

Figure 1: The questions in the original paper, source PDF page 8.

The source observes that compact and discrete groups have both requested properties. The result below proves that, within the entire LCA class, there are no other positive cases.

2 Proof intuition

Under the Fourier transform on an abelian group, convolution becomes multiplication. Thus a bounded function is a bounded L^2 -convolver exactly when its distributional Fourier transform is a bounded function. The problem becomes an uncertainty question: when can both a function and its Fourier transform be bounded without the original function lying in L^2 ?

The LCA structure theorem gives an open subgroup $\mathbb{R}^d \times K$ with K compact. If $d > 0$, a quadratic chirp has a bounded, unimodular Fourier transform and infinite L^2 norm. Multiplying its convolution operator by its adjoint produces the singular projection obtained by averaging over K ; the distance theorem in the source paper keeps that projection one unit away from every bounded-function convolver.

If $d = 0$, then K is compact open. Put different characters of K on different cosets of K . Orthogonality makes the Fourier transform bounded on disjoint fibers, although the kernel has infinite L^2 norm. The product with its adjoint is the indicator of the union of those fibers. Any uniform approximation to this projection would, after averaging along a dual fiber, give Fourier coefficients of an $L^1(K)$ function that remain close to 1 along infinitely many distinct characters. Riemann–Lebesgue forbids this.

3 Fourier formulation

Assume henceforth that G is LCA. Choose dual Haar measures so that $\mathcal{F} : L^2(G) \rightarrow L^2(\widehat{G})$ is unitary.

Lemma 1 (Multiplier criterion). *For $\varphi \in L^\infty(G)$, the following are equivalent:*

1. $\varphi \in \text{Conv}^\infty(G)$;
2. the distributional Fourier transform $\widehat{\varphi}$ is represented by some $m_\varphi \in L^\infty(\widehat{G})$.

In this case

$$\mathcal{F}\Lambda(\varphi)\mathcal{F}^{-1} = M_{m_\varphi}, \quad \|\Lambda(\varphi)\| = \|m_\varphi\|_\infty.$$

Moreover, if U is an open subgroup, extension by zero of a kernel on U corresponds to pulling its multiplier back by the restriction map $r : \widehat{G} \rightarrow \widehat{U}$.

Proof. A bounded translation-invariant operator on $L^2(G)$ is diagonalized by the Plancherel transform and hence is multiplication by an L^∞ function. Testing first on compactly supported Fourier test functions identifies this multiplier with the distributional Fourier transform of the convolution kernel. The converse follows from Plancherel and the distributional convolution identity. For an open subgroup, integration of the zero extension is integration over U , so the same test-function calculation gives pullback along r . $\square$

4 Classification theorem

Theorem 1. *Let G be a locally compact abelian group.*

1. $\text{Conv}^\infty(G) \subseteq L^2(G)$ if and only if G is compact or discrete.

2. The operator-norm closure

$$\mathcal{C}_G := \overline{\Lambda(\text{Conv}^\infty(G))}^{\|\cdot\|} \subseteq \text{VN}(G)$$

is a $*$ -subalgebra if and only if G is compact or discrete.

If G is neither compact nor discrete, the proof constructs $\varphi \in \text{Conv}^\infty(G) \setminus L^2(G)$ such that $\Lambda(\varphi)\Lambda(\varphi)^*$ has distance exactly 1 from $\Lambda(\text{Conv}^\infty(G))$ after normalizing $\|\Lambda(\varphi)\| = 1$.

Proof. The positive cases. If G is compact, Haar measure is finite, so $L^\infty(G) \subseteq L^2(G)$. Also $\text{Conv}^\infty(G) = L^\infty(G)$, and bounded functions are dense in $L^1(G)$; hence $\mathcal{C}_G = C_r^*(G)$.

If G is discrete and $\varphi \in \text{Conv}^\infty(G)$, then $\varphi = \Lambda(\varphi)\delta_e \in \ell^2(G)$. Conversely every element of $\text{VN}(G)$ is convolution by the ℓ^2 vector obtained by applying it to δ_e ; this vector is bounded. Therefore $\Lambda(\text{Conv}^\infty(G)) = \text{VN}(G)$. Both assertions follow in these two cases.

Structural reduction. The LCA structure theorem gives an open subgroup

$$U \cong \mathbb{R}^d \times K$$

with K compact [2]. Suppose that G is neither compact nor discrete. We split according to d .

Case 1: $d > 0$. Normalize Haar measure on K to have mass 1. On U set

$$\varphi_U(x, k) = e^{\pi i |x|^2},$$

and extend φ_U by zero to G . The Fresnel identity, in tempered distributions, gives

$$\widehat{e^{\pi i |\cdot|^2}}(\xi) = \zeta_d e^{-\pi i |\xi|^2}, \quad |\zeta_d| = 1.$$

Consequently, on $\widehat{U} \cong \mathbb{R}^d \times \widehat{K}$,

$$m_U(\xi, \gamma) = \zeta_d e^{-\pi i |\xi|^2} \mathbf{1}_{\{\widehat{1}_K\}}(\gamma).$$

Lemma 1 shows that the zero extension φ belongs to $\text{Conv}^\infty(G)$ with convolution norm 1. Since $|\varphi_U| = 1$ on a set of infinite Haar measure, $\varphi \notin L^2(G)$.

Let $P = \Lambda(\varphi)\Lambda(\varphi)^*$. Its multiplier is the pullback of $|m_U|^2$, so

$$P = \int_K \lambda(k) dk,$$

the orthogonal projection obtained by averaging over K . In particular $P \in \text{VN}_K(G)$ and $\|P\| = 1$. The subgroup K is closed and non-open, because $U/K \cong \mathbb{R}^d$. Every LCA group is amenable, so K admits a contractive approximate indicator in G . The distance theorem [1, Theorem 3.1] therefore yields

$$\text{dist}(P, \Lambda(\text{Conv}^\infty(G))) = \|P\| = 1.$$

Thus $P \notin \mathcal{C}_G$, although both factors defining P lie in $\Lambda(\text{Conv}^\infty(G))$. Hence $\mathcal{C}_G$ is not an algebra.

Case 2: $d = 0$. Now $U = K$ is compact open. Since G is nondiscrete, K is infinite; since G is noncompact, the discrete quotient G/K is infinite. Its annihilator

$$K^\perp = \{\chi \in \widehat{G} : \chi|_K = 1\}$$

is compact, and the restriction map

$$r : \widehat{G} \longrightarrow \widehat{K}$$

is onto, with fibers the cosets of $K^\perp$.

Choose distinct cosets $x_n + K$ in G/K and distinct characters $\gamma_n \in \widehat{K}$, for $n \geq 1$. Define a bounded measurable function by

$$\varphi(x_n + k) = \gamma_n(k) \quad (k \in K), \quad \varphi = 0 \quad \text{off} \quad \bigcup_{n \geq 1} (x_n + K).$$

Each occupied coset contributes 1 to the squared L^2 norm, and hence $\varphi \notin L^2(G)$.

For $N \geq 1$, let φ_N be the truncation to the first N occupied cosets. Then $\varphi_N \in L^1(G) \cap L^\infty(G)$, and character orthogonality on K gives

$$\widehat{\varphi_N}(\chi) = \begin{cases} \overline{\chi(x_n)}, & r(\chi) = \gamma_n \text{ for some } n \leq N, \\ 0, & r(\chi) \notin \{\gamma_n : 1 \leq n \leq N\}. \end{cases}$$

The displayed multipliers have pairwise disjoint fiber supports and converge pointwise to

$$m(\chi) = \begin{cases} \overline{\chi(x_n)}, & r(\chi) = \gamma_n \text{ for some } n, \\ 0, & r(\chi) \notin \{\gamma_n : n \geq 1\}. \end{cases}$$

In particular $m \in L^\infty(\widehat{G})$ and $\|m\|_\infty = 1$. If $f \in L^1(G) \cap L^2(G)$, dominated convergence gives $\varphi_N * f \rightarrow \varphi * f$ pointwise, while Plancherel and dominated convergence give

$$\varphi_N * f \longrightarrow \mathcal{F}^{-1}(m\widehat{f}) \quad \text{in } L^2(G).$$

After passage to an almost-everywhere convergent subsequence, the two limits agree. Hence convolution by φ extends to the multiplier M_m , and $\varphi \in \text{Conv}^\infty(G)$.

It remains to show that the closure is not an algebra. Set

$$E = r^{-1}(\{\gamma_n : n \geq 1\}).$$

Then the multiplier of $P := \Lambda(\varphi)\Lambda(\varphi)^*$ is $|m|^2 = \mathbf{1}_E$. We claim

$$\text{dist}(P, \Lambda(\text{Conv}^\infty(G))) = 1.$$

The upper bound follows by using the zero operator. For the lower bound, let $\theta \in \text{Conv}^\infty(G)$ have multiplier $a \in L^\infty(\widehat{G})$, and suppose for contradiction that

$$\|a - \mathbf{1}_E\|_\infty = \delta < 1.$$

Let μ be Haar probability measure on the compact subgroup $K^\perp$ and average along its cosets:

$$A = a * \mu.$$

Convolution by μ is contractive on L^∞ , and E is a union of $K^\perp$ -cosets, so

$$\|A - \mathbf{1}_E\|_\infty \leq \delta.$$

The inverse Fourier transform of μ is $\mathbf{1}_K$. Therefore A is the Fourier transform of $\theta \mathbf{1}_K$. Since $\theta \mathbf{1}_K \in L^1(K)$, A factors through $\widehat{K}$ and its values are the Fourier coefficients

$$A(\tilde{\gamma}) = \int_K \theta(k) \overline{\gamma(k)} dk, \quad r(\tilde{\gamma}) = \gamma.$$

By Riemann–Lebesgue these coefficients form an element of $c_0(\widehat{K})$. In particular their values along the distinct γ_n tend to zero. But (4) says that their absolute values are at least $1 - \delta > 0$ for every n , a contradiction. This proves (4), and again P is the product of two members of $\Lambda(\text{Conv}^\infty(G))$ but lies outside $\mathcal{C}_G$.

The two structural cases exhaust every LCA group that is neither compact nor discrete, proving both equivalences. $\square$

Remark 1 (Concrete compact-open example). *For $G = \mathbb{Z} \times \mathbb{T}$, the compact-open construction can be seen without any structure theory: $\varphi(n, t) = e^{2\pi i n t}$ is bounded and not in $L^2(G)$, while its Fourier transform on $\mathbb{T} \times \mathbb{Z}$ is again unimodular after exchanging the two variables.*

5 Verification and limitations

The proof was checked branch by branch against the source definitions. The main verifier targets were: (i) the finite-truncation and dominated-limit argument leading to (4); (ii) the identity turning dual-fiber averaging into multiplication by $\mathbf{1}_K$; and (iii) the hypotheses of the source distance theorem in Case 1. All three checks pass. No computation is used as proof.

This result is complete only in the LCA subclass. It does not classify arbitrary locally compact groups, and it does not give a full intrinsic description of $\mathcal{C}_G$ in the negative cases. A bounded local-corpus and web/arXiv search found the source paper and general literature on convolvers and pseudomeasures, but no paper stating these two LCA equivalences. Because the ingredients are classical and older literature may use different terminology, novelty confidence is moderate rather than high.

6 Human review recommendation

Send to a harmonic analyst for review. The proof is short enough to audit directly. The compact-open construction and the fiber-average Riemann–Lebesgue obstruction deserve the closest check; bibliographic review should search older pseudomeasure terminology before a public novelty claim.

References

- [1] B. E. Forrest, N. Spronk, and P. J. Wood, *Operator Segal algebras in Fourier algebras*, *Studia Math.* **179** (2007), no. 3, 277–295; arXiv:math/0601509.
- [2] E. Hewitt and K. A. Ross, *Abstract Harmonic Analysis, Vol. I*, 2nd ed., Springer, 1979.